FACE RECOGNITION ACCESS CONTROL SYSTEM

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PRESENTED BY :

Anastasiia Teodorovska

Mariana Pashchuk

THE DEVELOPED SYSTEM WAS INSPIRED BY A SECURITY SCENARIO IN WHICH ACCESS TO A RESTRICTED LABORATORY IS GRANTED ONLY TO CAPTAIN DEMMING THROUGH FACIAL VERIFICATION.

2026

PYTHON PROGRAMMING PROJECT

WHY DO WE NEED FACIAL RECOGNITION?

TRADITIONAL METHODS PROBLEMS

PASSWORDS CAN BE FORGOTTEN

ACCESS CARDS CAN BE LOST

CREDENTIALS CAN BE STOLEN OR SHARED

FACIAL RECOGNITION SOLUTION

BIOMETRIC VERIFICATION USING FACIAL FEATURES

ENHANCED SECURITY AND RELIABILITY

FAST AND CONTACTLESS AUTHENTICATION

GOAL

Develop an intelligent face-based access control system that can automatically recognize an authorized user and decide whether access should be granted or denied.

PROJECT OBJECTIVES

The development of the Face Recognition Access Control System was divided into several key stages:

1.

FACE DETECTION

Detect and locate human faces in images captured by the webcam. The system identifies facial regions and extracts them for further processing and recognition.

2.

DATASET CREATION

Collect and store multiple facial images of the authorized user under different expressions and viewing conditions to create a reliable training dataset.

3.

MODEL TRAINING

Train the LBPH face recognition model using the collected dataset. During this stage, facial features are analyzed and transformed into a trained recognition model.

4.

USER IDENTIFICATION

Recognize and verify the identity of a detected person by comparing facial features with the data stored in the trained model.

5.

ACCESS DECISION

Automatically grant or deny access based on the recognition confidence score and predefined verification criteria.

6.

ACTIVITY LOGGING

Record access attempts, recognition results, timestamps, and system decisions to ensure monitoring, traceability, and security auditing.

PROJECT WORKFLOW

The system follows a sequential workflow that processes webcam input and ultimately produces an access control decision.

CAMERA INPUT

The webcam captures a real-time image or video frame.

FACE DETECTION

The system detects the presence and location of a human face.

IMAGE CAPTURE

Detected facial regions are cropped and prepared for processing.

DATASET CREATION

Captured face images are stored and organized for training.

LBPH MODEL TRAINING

The LBPH algorithm learns facial patterns from the dataset.

FACE RECOGNITION

The trained model compares a detected face with stored facial data.

ACCESS DECISION

The system grants or denies access based on the recognition result.

WHAT IS FACE RECOGNITION?

DEFINITION

Face Recognition is a computer vision technology that automatically identifies or verifies a person based on unique facial characteristics.

Unlike traditional authentication methods, such as passwords or access cards, facial recognition uses biometric information that is unique to each individual.

KEY FACIAL FEATURES ANALYZED

- MOUTH
- EYES
- NOSE
- FACIAL GEOMETRY & PROPORTIONS

These characteristics are transformed into a numerical representation that can be compared with previously stored facial data to determine a person's identity.

FACE DETECTION VS FACE RECOGNITION

Although these terms are often used interchangeably, they represent two different stages of a facial recognition system.

FACE DETECTION

The system detects a face and draws a bounding box around it.

← **EXAMPLE** →

FACE RECOGNITION

The system compares the detected face with stored facial data and identifies the person.

1.

FACE DETECTION

- Detects the presence of a face in an image
- Determines the face location
- Answers: "Is there a face?"
- First stage of the system

2.

FACE RECOGNITION

- Identifies whose face has been detected
- Determines the person's identity
- Answers: "Whose face is it?"
- Second stage of the system

TECHNOLOGIES USED

IMPLEMENTATION TECHNOLOGIES

The Face Recognition Access Control System was developed using Python and several supporting libraries that provide functionality for image processing, face recognition, data management, and system operations.

WHY THESE TECHNOLOGIES?

The selected technologies provide an efficient combination of computer vision, data processing, and system management tools required to implement a real-time face recognition access control system.

PYTHON

Core programming language used for system development

OPENCV

Face detection, image processing, and webcam interaction

OS

File and directory management

DATETIME

Recording timestamps for access logs

YAML

Storage of configuration parameters and system settings

PYGAME

Audio notifications and system feedback

NUMPY

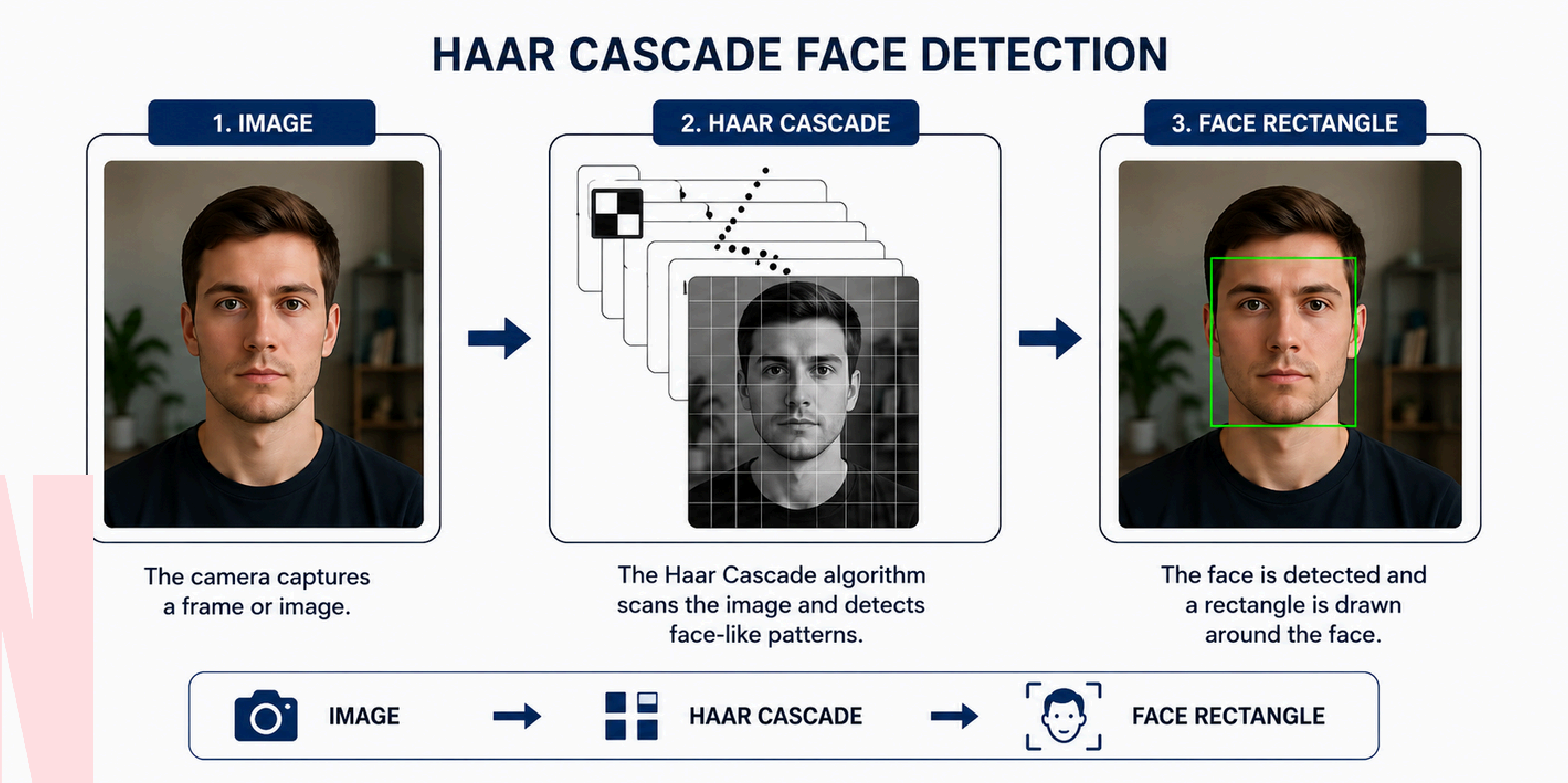
Numerical operations and image data manipulation

HAAR CASCADE FACE DETECTION

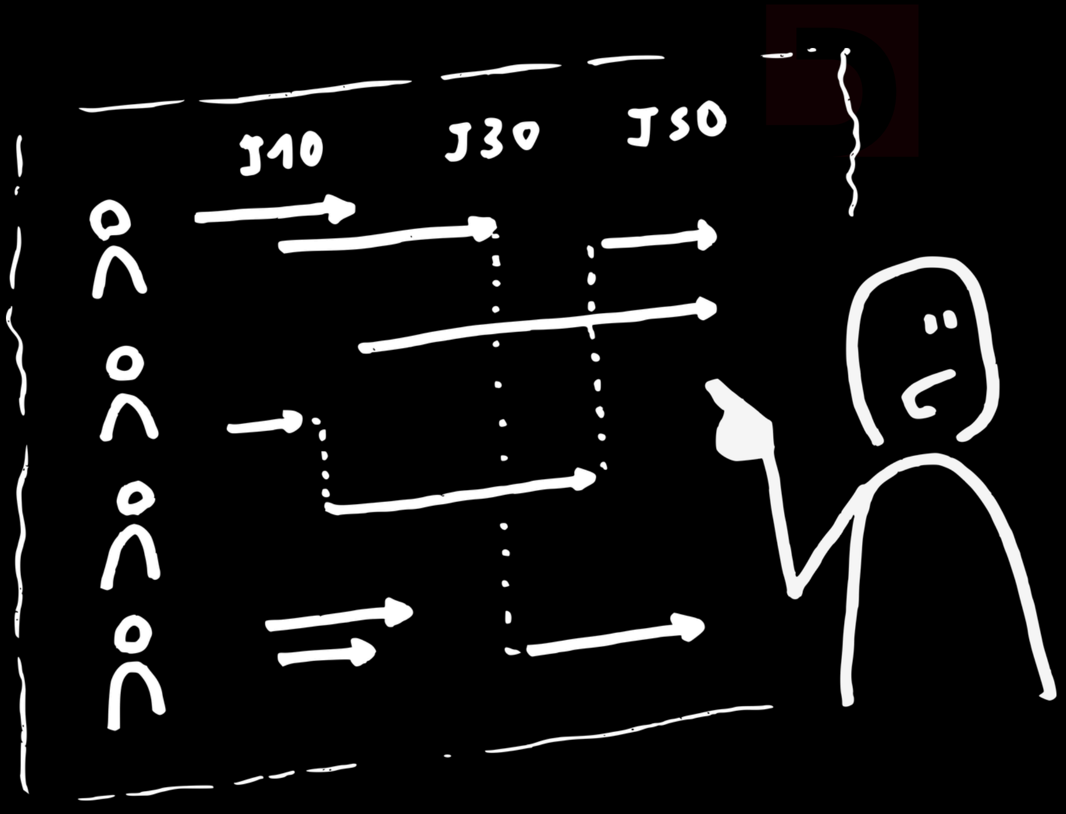
Before a person can be recognized, the system must first detect the presence and location of a face within an image.

In this project, face detection is performed using the Haar Cascade algorithm provided by OpenCV. The algorithm scans the image and searches for facial patterns that correspond to human faces.

Once a face is detected, the system draws a bounding rectangle around it and extracts the facial region for further processing.



LOCAL BINARY PATTERN HISTOGRAM (LBPH)



DEFINITION

The Local Binary Pattern Histogram (LBPH) algorithm is a classical face recognition method used to identify individuals based on facial texture patterns.

LBPH analyzes the local characteristics of a face, converts them into numerical patterns, and creates a unique representation that can be compared with previously trained facial data.

PROJECT APPLICATION

LBPH is used as the core recognition algorithm responsible for identifying authorized users and supporting access control decisions.

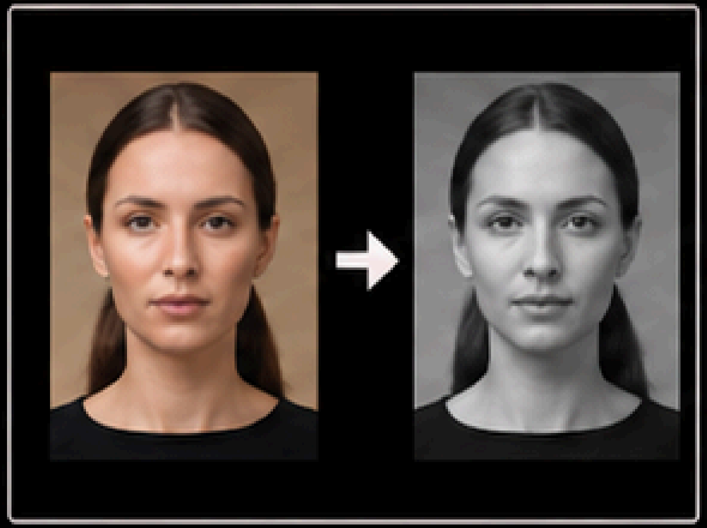
KEY ADVANTAGES

- Fast and computationally efficient
- Simple to implement and train
- Suitable for real-time applications
- Robust to illumination and lighting changes
- Provides reliable recognition in controlled environments

HOW LBPH WORKS

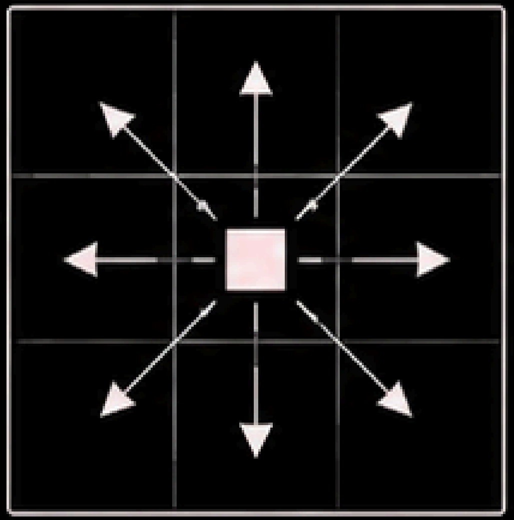
THE LOCAL BINARY PATTERN HISTOGRAM (LBPH) ALGORITHM RECOGNIZES FACES BY ANALYZING LOCAL TEXTURE INFORMATION AND TRANSFORMING IT INTO A NUMERICAL REPRESENTATION.

1 CONVERT IMAGE TO GRAYSCALE



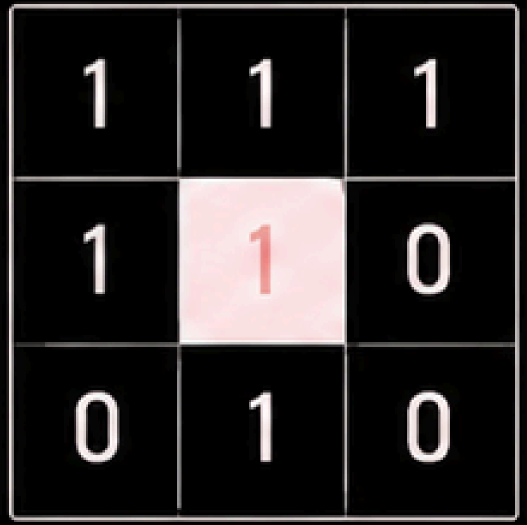
The input image is converted to grayscale to simplify processing and reduce computational complexity.

2 COMPARE NEIGHBORING PIXELS



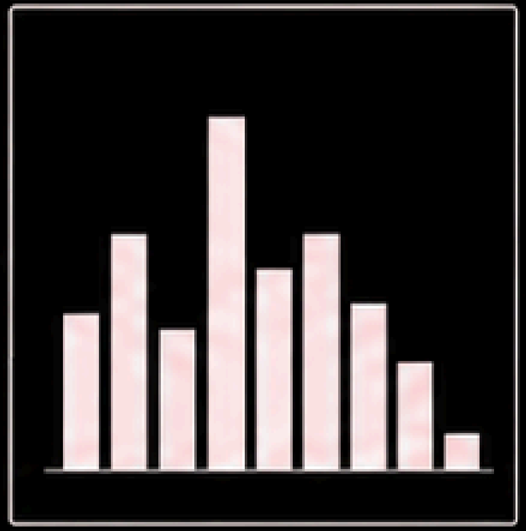
Each pixel is compared with its surrounding neighboring pixels.

3 CREATE A BINARY PATTERN



If a neighboring pixel has a value greater than or equal to the center pixel, it is assigned a value of 1; otherwise, it is assigned 0. This produces a binary pattern.

4 BUILD HISTOGRAMS

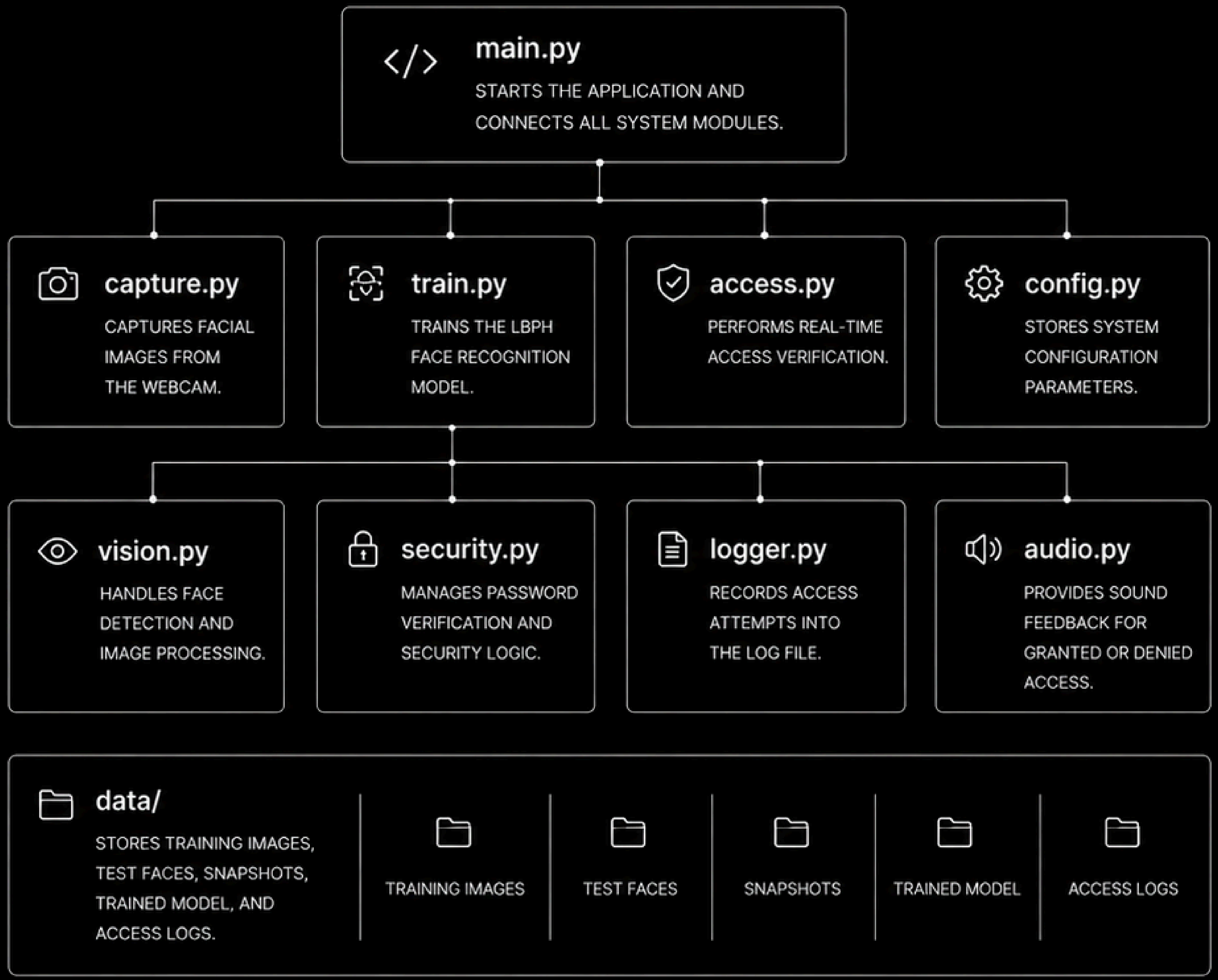


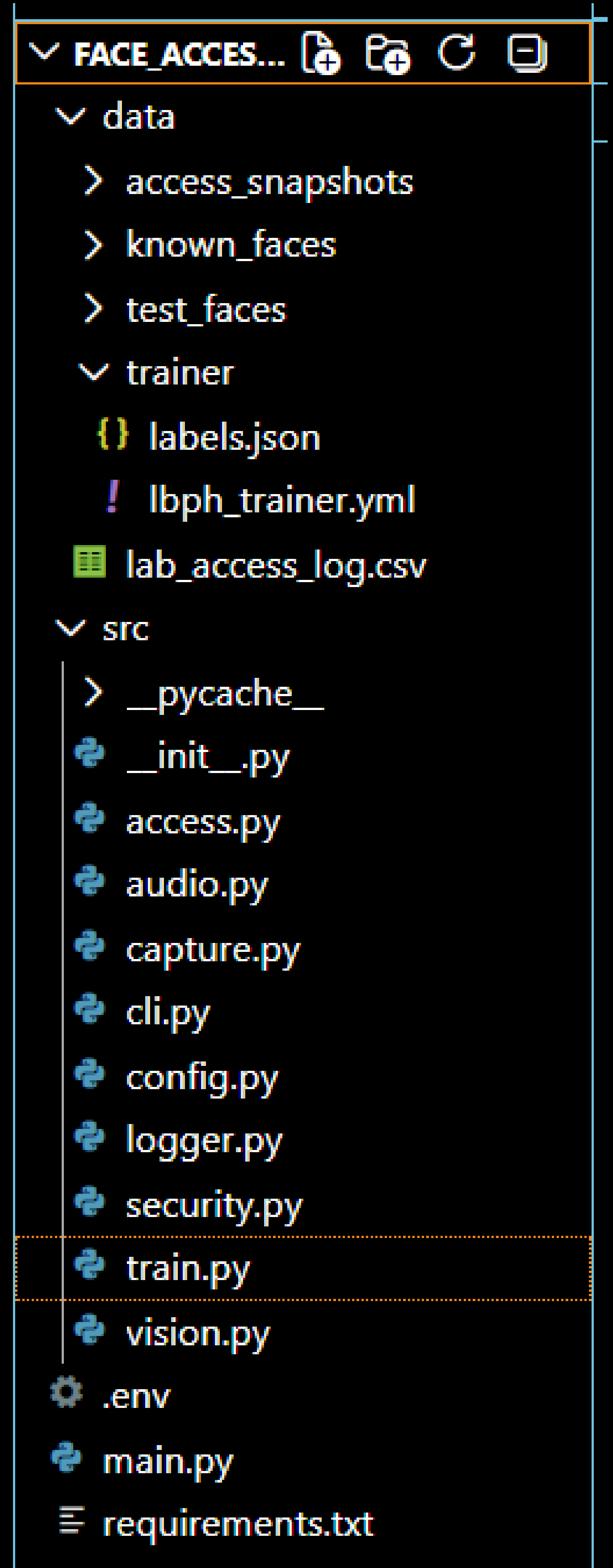
The generated binary patterns are grouped into histograms that describe local facial texture characteristics.

5 COMPARE WITH TRAINED FACES

The resulting histograms are compared with previously stored facial representations to identify the most similar face.

SYSTEM ARCHITECTURE





The project is organized into separate modules, where each file is responsible for a specific part of the face recognition access control system. This approach enhances code organization and simplifies maintenance.

DATASET CREATION

The first stage of the system is the creation of a facial image dataset used for training the recognition model.

During data collection, the webcam continuously captures video frames while the Haar Cascade detector locates the user's face in real time.

Multiple facial images are automatically captured under different positions and slight variations in appearance to improve recognition accuracy.

All collected images are stored in the dataset directory and later used to train the LBPH face recognition model.

DATASET CREATION PROCESS

WEBCAM ACTIVATION
Starts video capture session

↓

FACE DETECTION
Locates the face using Haar Cascade

↓

IMAGE CAPTURE
Collects multiple facial images

↓

DATASET STORAGE
Automatically saves captured samples

↓

LBPH TRAINING DATA
Provides images for model training

MODEL TRAINING

After the dataset is created, the system proceeds to the training stage, where the face recognition model is generated.

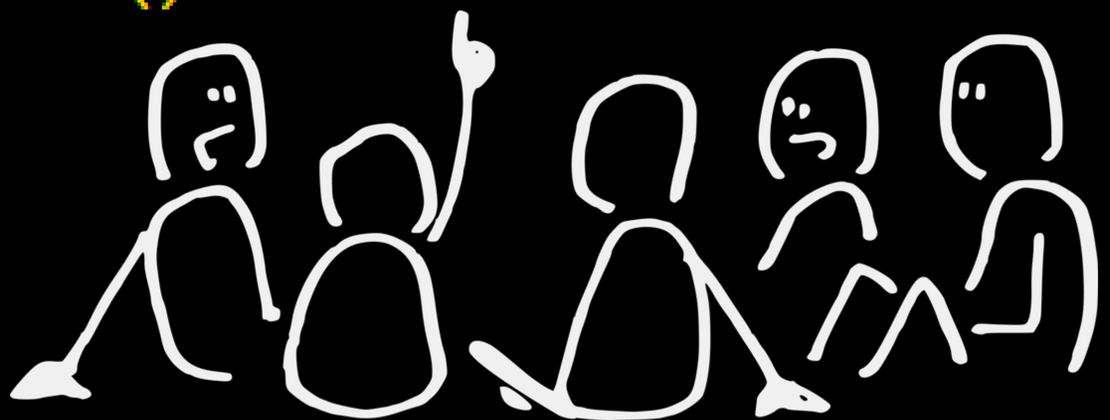
The training process reads all facial images, extracts the detected face regions, and associates each image with the corresponding user label.

Using this information, the LBPH algorithm learns unique facial patterns and creates a recognition model capable of identifying authorized users.

The trained model is then saved and later used during real-time authentication.

KEY OPERATIONS

```
recognizer = cv.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
recognizer.write(str(MODEL_PATH))
```



TRAINING PROCESS

DATASET IMAGES
Load captured facial samples

↓

FACE EXTRACTION
Extract detected face regions

↓

LABEL GENERATION
Assign user identifiers

↓

LBPH TRAINING
Learn facial patterns

↓

TRAINED MODEL
Save recognition model

ACCESS VERIFICATION

Once the LBPH model has been trained, the system is ready to perform real-time user verification and access control.

When a person appears in front of the webcam, the system detects the face and extracts the facial region for analysis. The trained LBPH model then compares the detected face with the stored biometric profile.

Based on the similarity score obtained during comparison, the system evaluates whether the user matches the authorized identity.

The final access decision is made automatically, resulting in either access being granted or denied.

VERIFICATION PROCESS

CAMERA INPUT
Capture live facial image

FACE DETECTION
Locate the face using Haar Cascade

FACE RECOGNITION
Compare with stored profile

CONFIDENCE EVALUATION
Assess similarity score

ACCESS DECISION
Grant or deny access

USER GUIDE

THE DEVELOPED FACE RECOGNITION ACCESS CONTROL SYSTEM IS DESIGNED TO PROVIDE SECURE AND INTUITIVE ACCESS VERIFICATION THROUGH A COMBINATION OF PASSWORD AUTHENTICATION AND BIOMETRIC FACE RECOGNITION.

1.

CREATE A FACIAL DATASET

Launch the image capture module and collect multiple facial images using the webcam. The system automatically detects the user's face and stores the captured images for model training.

2.

TRAIN THE RECOGNITION MODEL

Run the training module to generate an LBPH face recognition model from the collected dataset.

3.

START THE ACCESS CONTROL SYSTEM

Launch the access verification module to begin the authentication process. The system loads the trained model and prepares user verification.

4.

ENTER THE ACCESS PASSWORD

The user is prompted to enter the predefined access password. Only users with a valid password can proceed to biometric verification.

5.

PERFORM FACIAL VERIFICATION

After successful password validation, the webcam is activated and the system captures multiple verification images. The detected face is analyzed and compared with the trained facial model.

6.

ACCESS GRANTED OR DENIED

Based on the facial recognition result and confidence evaluation, the system automatically grants or denies access.

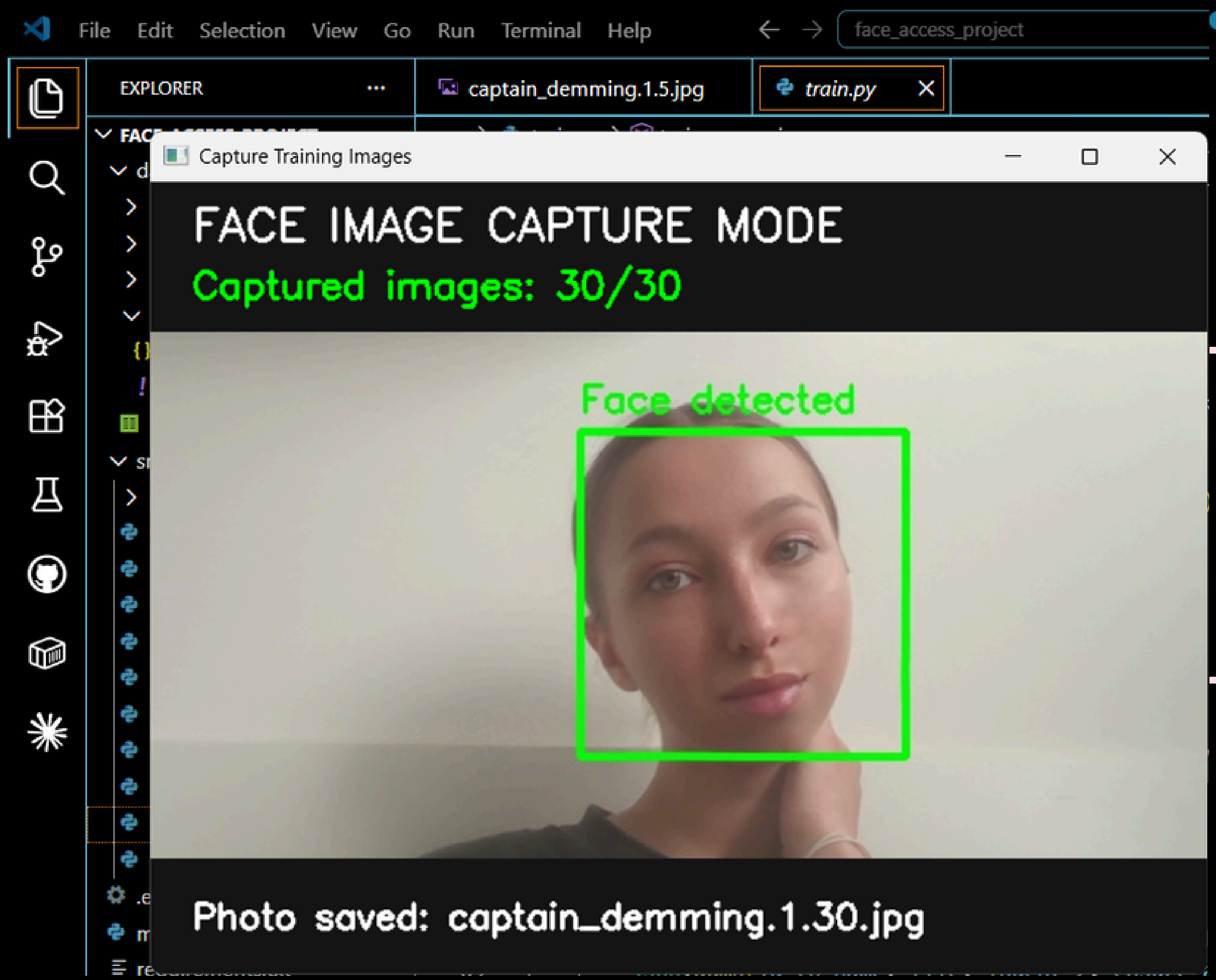
7.

REVIEW ACCESS LOGS

All verification attempts, including successful and unsuccessful access requests, are recorded and can be reviewed for monitoring and security purposes.

DEMONSTRATION OF THE DEVELOPED SYSTEM

DATASET COLLECTION + TRAINING THE LBPH MODEL




```
PS C:\Users\Admin\Desktop\face_access_project> python main.py capture --name "Captain Demming" --user-id 1 --samples 30
Face capture mode started.
Please look directly at the camera.
The system will take several photos for training.
Slowly move your head left and right.
Try a neutral, happy, and serious expression.
Keep your face inside the green rectangle.
```

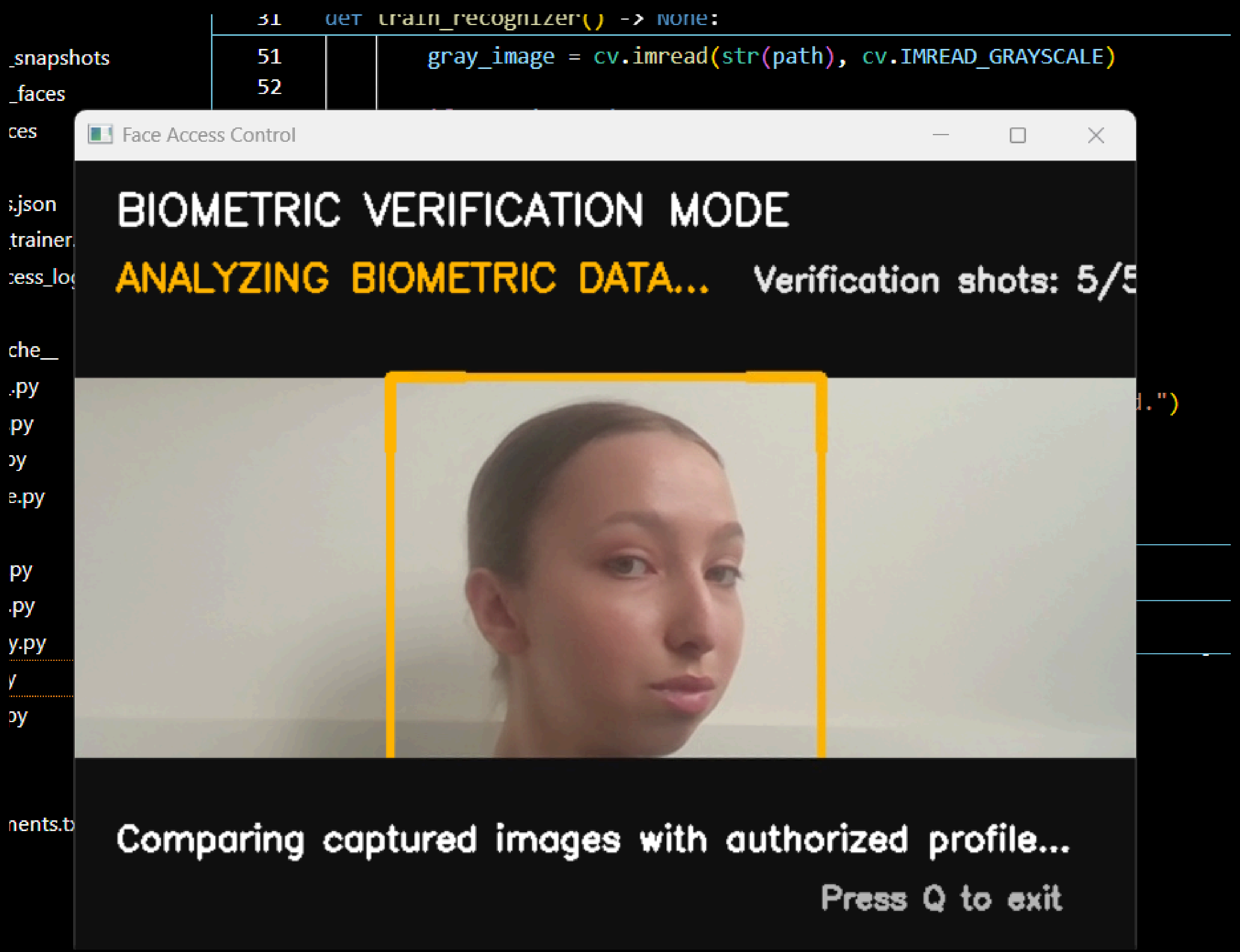
```
Image capture completed.
30 training images were saved.
Saved 30 image(s) to C:\Users\Admin\Desktop\face_access_project\data\known_faces
```

```
PS C:\Users\Admin\Desktop\face_access_project> python main.py train
Training complete.
Model saved to: C:\Users\Admin\Desktop\face_access_project\data\trainer\lbph_trainer.yml
Labels saved to: C:\Users\Admin\Desktop\face_access_project\data\trainer\labels.json
```

The system captures multiple facial images of Captain Demming through a webcam to create a training dataset for the recognition model. The collected images of Captain Demming are processed and used to train the LBPH face recognition model.

DEMONSTRATION OF THE DEVELOPED SYSTEM

REAL-TIME FACE RECOGNITION + ACCESS GRANTED



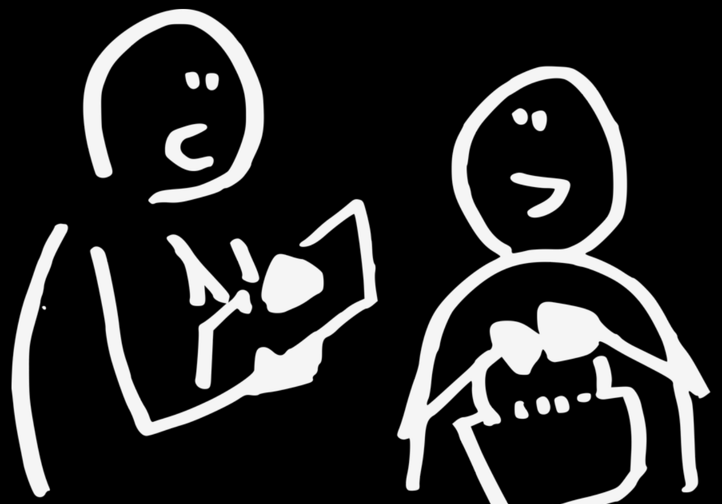

```
PS C:\Users\Admin\Desktop\face_access_project> python main.py access
============================================================
LABORATORY SECURITY SYSTEM
============================================================
Laboratory security system activated.
Please enter the access password.
Enter access password:
Password accepted.
Look at the camera and place your face inside the frame.
Face detected. Verification started.
[SHOT 1] Predicted ID: 1, Name: Captain Demming, Distance: 50.04, Accepted: True
[SHOT 2] Predicted ID: 1, Name: Captain Demming, Distance: 44.15, Accepted: True
[SHOT 3] Predicted ID: 1, Name: Captain Demming, Distance: 33.13, Accepted: True
[SHOT 4] Predicted ID: 1, Name: Captain Demming, Distance: 65.74, Accepted: True
[SHOT 5] Predicted ID: 1, Name: Captain Demming, Distance: 63.07, Accepted: True
Analyzing biometric data.
Identity confirmed.
Access granted. Welcome, Captain Demming.
```

The system detects a face, extracts facial features, and compares them with the trained model of Captain Demming. When the detected face matches Captain Demming's facial profile with sufficient confidence, access is granted automatically.

CONCLUSIONS

The Face Recognition Access Control System was successfully designed, implemented, and tested using Python, OpenCV, and the LBPH face recognition algorithm.

The project combines biometric authentication, password verification, and automated access control into a single security solution capable of operating in real time.



FINAL OUTCOME

The developed system demonstrates the practical application of computer vision technologies for secure user authentication and intelligent access management.

ACHIEVEMENTS

- Face detection implemented using Haar Cascade
- Facial dataset successfully created and processed
- LBPH recognition model successfully trained
- Real-time face recognition achieved
- Password-based authentication integrated
- Automated access control implemented
- Access logging and monitoring functionality added

THANK YOU

THANK YOU

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